1. Master Time Value of Money (TVM) & Profit-Test Architecture — Humana NPV Model (v4.2.0)

Overview: Why Monthly Intramonth Timing Matters

The Humana NPV Forecasting Model (v4.2.0) values blocks of business across two core actuarial lenses:

1. **Cash NPV (Pre-Tax Operational View):** Pre-tax cash flows timed at incurred dates.
2. **Distributable Earnings (DE) NPV (Corporate Decision Measure):** After-tax statutory distributable earnings accounting for chain-ladder claim payment lags, IBNR claim reserves, MLR remittance liabilities, required Risk-Based Capital (RBC), investment income on backing assets, federal income tax (21%), and the IRC §848 Deferred Acquisition Cost (DAC) tax.

In Version 1 of the model, cash flows were projected annually; an event-level backtest proved that annual conventions biased Medicaid NPV by -12%. Version 4.2 projects month-by-month (k = 0, 1, ..., 12Y - 1) using exact intramonth timing and compound monthly decrement conversions:

$$q_{monthly} = 1 - (1 - q_{annual})^{(1/12)}$$

so that twelve months of compounding reproduce the annual decrement rate exactly.

Intramonth Actuarial Timing & Discounting Architecture (Month k)

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                    INTRAMONTH ACTUARIAL TIMING & DISCOUNTING ARCHITECTURE (MONTH k)
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  TIME 0 (Issue / Go-Live)       MONTH START (t = k/12)       MID-MONTH (t =
(k+0.5)/12)      MONTH END (t = (k+1)/12)
  ------------------------       ----------------------
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  * Acquisition Cost (MS, DV)    * In-Force Count: IF_start   * Deaths occur: 0.5 *
Deaths    * Lapses take effect
    [-Cohort * AcqCost]          * Premium Billed (MS, DV)      -> Exposure = IF -
0.5*Deaths   (member covered all mo)
  * Implementation / Start-up    * Commission Paid (%)        * Incurred Claims (MS,
DV, MD)  * Statutory Reserves &
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    [-MD_Startup]                * Premium Tax Paid (%)       * Maintenance / Admin
Paid        Required Capital held
                                                              * Medicaid Capitation &
* Distributable Earnings
                                                                Joiner/Leaver Churn
(DE) released

* Tax & DAC Tax (Year-End)

  DISCOUNT FACTORS (v = 1 / (1 + DiscRate), where DiscRate = 8.5% hurdle rate):
  * Time 0:      1.0000          * Start: v^(k/12)            * Mid: v^((k + 0.5)/12)
* End: v^((k + 1)/12)
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\Exact Discount Factor Equations

Let DiscRate be the annual corporate hurdle rate (8.5% in the base assumption set), and let v = 1 ÷ (1 + DiscRate). For month index k in 0, 1, ..., 12Y - 1:

- **Issue / Pre-Go-Live (t = 0):** DF_0 = 1.0000
- **Start of Month k (t = k ÷ 12):** DF_start(k) = (1 + DiscRate)^(-k ÷ 12)
- **Mid-Month k (t = (k + 0.5) ÷ 12):** DF_mid(k) = (1 + DiscRate)^(-(k + 0.5) ÷ 12)
- **End of Month k (t = (k + 1) ÷ 12):** DF_end(k) = (1 + DiscRate)^(-(k + 1) ÷ 12)
- **Medicaid Annual Settlement (6 months after contract year y closes, t = y + 0.5):** DF_settle(y) = (1 + DiscRate)^(-(y + 0.5))

Lens 1: Cash NPV (Pre-Tax Operational View)

Cohort Lines (Medicare Supplement and Individual Dental & Vision)

- **Start-of-Month Net Cash Flow:** StartCash(k) = Premium(k) - Commission(k) - PremTax(k)
- **Mid-Month Net Outflow:** MidOutflow(k) = IncurredClaims(k) + Maintenance(k)
- **Monthly Present Value:** PV_Cash(k) = [Premium(k) - Commission(k) - PremTax(k)] × (1 + DiscRate)^(-k ÷ 12) - [IncurredClaims(k) + Maintenance(k)] × (1 + DiscRate)^(-(k + 0.5) ÷ 12)
- **Total Cash NPV:** Cash_NPV = SUM over k = 0..(12Y - 1) of [PV_Cash(k)] - (Cohort × AcqCost_per_policy)

Medicaid Managed Care (Open Contract)

- **Monthly Operating Present Value (discounted mid-month):** `PV_Op(k) = [Revenue(k) - Medical(k) - PremTax(k) - Admin(k)] × (1 + DiscRate)^(-(k + 0.5) ÷ 12)`
- **Contract-Year Expected Present Value (weighted by renewal probability P_inforce(y), where P_inforce(y) = 1.0 for y ≤ Term, and MD_RenewProb for y > Term):** `PV_Year(y) = P_inforce(y) × [ SUM over k in Year y of PV_Op(k) - NetSettlement(y) × (1 + DiscRate)^(-(y + 0.5)) ]`
- **Total Contract Cash NPV:** `Cash_NPV = SUM over y = 1..Y of [ PV_Year(y) ] - MD_Startup`

Lens 2: Distributable Earnings (DE) Profit Test (*_Profit Sheets)

The Profit Test extends the projection horizon by a 9-month claim runoff period ($H = 12Y + 9 \text{ months}$) so that all incurred claims and lagged settlements run off completely to zero.

Step 1: Chain-Ladder Claim Payment Lag & Statutory Claim Reserve

From SQL study `03_completion_factors.sql`, each line has a 9-lag monthly payment pattern Pay_0, Pay_1, ..., Pay_8 summing to 100%:

- **Paid Claims in Month k:** `PaidClaims(k) = SUM over lag j = 0..min(k, 8) of [ Pay_j × IncurredClaims(k - j) ]`
- **Statutory Claim Reserve (IBNR) at End of Month k:** `ClaimReserve(k) = ClaimReserve(k - 1) + IncurredClaims(k) - PaidClaims(k)  (with ClaimReserve(-1) = 0)`
- **Change in Claim Reserve:** $\Delta\text{Reserve}(k) = \text{ClaimReserve}(k) - \text{ClaimReserve}(k-1) = \text{IncurredClaims}(k) - \text{PaidClaims}(k)$

Step 2: Required Risk-Based Capital (RBC Proxy)

Required capital is held at the end of each month as a percentage (RC_{pct}) of annualized paid claims, tapering naturally as claims run off:

- **Required Capital at End of Month k:** $\text{ReqCapital}(k) = \text{RC_pct} \times 12 \times \text{PaidClaims}(k)$
- **Change in Required Capital:** $\Delta\text{Capital}(k) = \text{ReqCapital}(k) - \text{ReqCapital}(k-1)$

Step 3: Investment Income on Backing Assets

`Assets backing statutory reserves and required capital earn the annual investment earned rate PT_Earn (converted to monthly compound rate j = (1 + PT_Earn)^(1/12) - 1):`

- **Investment Income on Reserves (Month k):** `InvInc_Res(k) = j × [ ClaimReserve(k - 1) + MLR_Liability(k - 1) ]`
- **Investment Income on Required Capital (Month k):** `InvInc_Cap(k) = j × ReqCapital(k - 1)`

Step 4: Pre-Tax Statutory Book Profit

Start-of-month cash earns a full month of interest (1 + j) to month-end; mid-month cash net of paid claims earns a half-month of interest (1 + j)^0.5 to month-end:

- **Cohort Book Profit (Med Supp & D&V):** `BookProfit(k) = StartCash(k) × (1 + j) + [MidCash_excl_claims(k) - PaidClaims(k)] × (1 + j)^0.5 - [ClaimReserve(k) - ClaimReserve(k - 1)] + InvInc_Res(k)`
- **Medicaid Book Profit:** `BookProfit(k) = [MidCash_excl_medical(k) - PaidClaims(k)] × (1 + j)^0.5 - [ClaimReserve(k) - ClaimReserve(k - 1)] - MLR_Accrued(k) + InvInc_Res(k)`

Step 5: Federal Income Tax & Distributable Earnings (Before DAC Tax)

- **Federal Income Tax (PT_Tax = 21%):** FedTax(k) = 21% × [BookProfit(k) + InvInc_Cap(k)]
- **Monthly Distributable Earnings (DE):** `DE(k) = BookProfit(k) + InvInc_Cap(k) - FedTax(k) - [ReqCapital(k) - ReqCapital(k - 1)]`

Step 6: IRC §848 Deferred Acquisition Cost (DAC) Proxy Tax (Med Supp)

Under IRC §848, guaranteed-renewable Accident & Health contracts must capitalize a statutory percentage of net premiums (MS_DAC = 9.2% for Med Supp) as "specified policy acquisition expenses" and amortize them over 180 months (15 years) using the half-year convention:

- **DAC Capitalized in Tax Year t (t = 1..20):** DAC_Cap(t) = 9.2% × NetPremiums(t)
- **15-Year Half-Year Amortization Weights (W_m for lag m = t - y in 0..15):** W_0 = 1/30 (Year 1: 6 months ÷ 180), W_1 = ... = W_14 = 1/15 (Years 2..15: 12 months ÷ 180), W_15 = 1/30 (Year 16: remaining 6 months ÷ 180)
- **DAC Amortized in Tax Year t (t = 1..35):** `DAC_Amort(t) = SUM over policy years y = max(1, t - 15)..min(t, Y) of [ DAC_Cap(y) × W_(t - y) ]`
- **Extra Tax Due to IRC §848 in Tax Year t:** `ExtraTax_DAC(t) = 21% × [ DAC_Cap(t) - DAC_Amort(t) ]`
- **Present Value of DAC Tax Drag:** `PV_DAC_Tax = SUM over t = 1..(Y + 15) of [ ExtraTax_DAC(t) × (1 + DiscRate)^(-t) ]`

Step 7: Final Distributable Earnings NPV (DE NPV) & Accounting Identity Check

- **Final DE NPV Formula:** `DE_NPV = -Cost_0 × (1 - 21%) + SUM over k = 0..(12Y + 8) of [ DE(k) × (1 + DiscRate)^(-(k + 1) ÷ 12) ] - PV_DAC_Tax`
- **Built-in Profit-Test Identity Control:** When PT_Tax = 0%, RC_pct = 0%, MS_DAC = 0%, PT_Earn = DiscRate, and Pay_0 = 100% (immediate claim payment), DE_NPV equals Cash_NPV to the exact cent.

Step 8 (v4.2): Risk-Based Pricing Solve on DE NPV

- **Pricing Target:** DE NPV is the decision measure the pricing solve targets. Each line's premium is solved so that P(DE_NPV < 0) across the 10,000-scenario Monte Carlo (parameter risk + process risk + rate-filing response, common random numbers) is at most the risk tolerance: equivalently, the 10th percentile of DE_NPV is solved to zero with Brent's method (04_npv_engine.py, pricing_solve()).
- **Why Not Break-Even:** Pricing to DE_NPV = 0 leaves about a one-in-two chance of missing the 8.5% hurdle. The risk-based target sizes the margin above break-even by each line's own risk. Trend stays at its best estimate; the margin protects against it, it does not replace it.
- **Governance & Constraints:** The 10% tolerance is corporate pricing policy (judgment), held in config/pricing_targets.json and covered by the assumption approval hash (AS-SYN-2026-009). Regulatory limits are checked, never relaxed: the 65% Med Supp minimum loss ratio and state-set Medicaid capitation. At the recommended premiums the independent Excel model still ties to Python to the cent (tests/e2e_excel_pricing.py).